THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 3

Chapter 11: **Electricity** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A metal wire carries 2 A. Heat produced in 10 s is 400 J. If the current is raised to 4 A for 5 s, the heat produced is 1

(a) 200 J **(b)** 400 J **(c)** 800 J **(d)** 1600 J

2. Assertion (A): Electric current is a scalar quantity. Reason (R): Currents add arithmetically at a junction and do not follow the rules of vector addition. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

3. An electric kettle rated 1.1 kW at 220 V is connected to a 110 V supply. The power drawn is nearly 1

(a) 2200 W **(b)** 550 W **(c)** 275 W **(d)** 1100 W

4. The kilowatt hour used by a 250 W lamp for 4 hours is 1

(a) 250 kWh **(b)** 0.25 kWh **(c)** 4 kWh **(d)** 1 kWh

5. Assertion (A): The bulb in a flashlight glows immediately on closing the switch even though electrons drift slowly. Reason (R): The electric field is set up in the conductor almost instantly. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. The ratio of resistances of 60 W and 100 W bulbs of the same voltage rating is 1

(a) 6 : 10 **(b)** 5 : 3 **(c)** 3 : 5 **(d)** 1 : 1

7. Which of these changes will NOT increase the resistance of a copper wire? 1

(a) Doubling the length **(b)** Halving the area
(c) Reducing the temperature **(d)** Replacing with nichrome

8. Assertion (A): A 60 W bulb has a higher resistance than a 100 W bulb for the same voltage rating. Reason (R): Resistance is given by V squared divided by power for a rated bulb. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

9. The current in a resistor R is doubled by changing the voltage. The power dissipated becomes 1

(a) four times **(b)** the same **(c)** twice **(d)** half

10. A fuse of rating 3 A is used in a circuit with appliances of 220 V. The total power which can be used is 1

(a) 73 W **(b)** 660 W **(c)** 1100 W **(d)** 220 W

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is the power rating of a fuse specified as a current rating? 2

12. Two resistors, when connected in series, have a total of 25 ohm and in parallel 4 ohm. Find the resistors. 2

13. A bulb rated 60 W, 220 V is connected to 110 V. Find the current drawn. 2

14. What is the effect on the heat produced in a wire if both the current and the resistance are doubled for the same time? 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. A cell sends 0.5 A through a 4 ohm resistor and 0.25 A through an 8 ohm resistor. Find the emf and the internal resistance. 3

16. The resistance between points A and B of a 12 ohm wire bent into a square is calculated between adjacent corners. Find it. 3

17. A nichrome wire and a copper wire of the same dimensions are connected in series. Which gets hotter and why? 3

OR

A hot plate with a resistance of 55 ohm works on 220 V. Find the current, the power and the cost of using it for 5 hours at Rs 5 per unit. 3

18. How will the brightness of a bulb change if one more identical bulb is connected in series with it? Explain. 3

OR

The current of 2 A flows through the resistor of 10 ohm. How much charge flows and how much energy is dissipated in 30 s? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Establish the relation between the heat produced, the current, the resistance and the time. A nichrome coil produces 8.4×10^5 J in 10 minutes from a 220 V supply. Find its resistance and the current. 5

OR

Compare the series and parallel arrangements of two identical bulbs on a battery with reference to the brightness, the total power and the effect of the failure of one bulb. Support with a numerical example for two 10 ohm bulbs on 20 V. 5

20. A student has resistors of 3 ohm, 6 ohm and 9 ohm. Show how she can obtain 5

(a) the highest possible resistance

(b) the lowest possible resistance

(c) 11 ohm. Verify with calculations.

OR

A power station delivers 22 kW to a village through a line of total resistance 2 ohm. Compare the power lost in the line if the power is sent at 220 V and at 2200 V. 5

END OF QUESTION PAPER Total: 40 Marks